VisionCam

-AI Driver Alert System

Team-Calm and Cool

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Problem Addressed

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“A second of distraction, and a life could be lost!!”

- Thousands of people are injured or killed each year due to drivers falling asleep, not paying attention to the road, or looking at their phones rather than their driving.

Gaps in Current Solutions

- **Current systems are expensive and limited to premium vehicles.**
- **They detect sleepiness only, missing distraction and phone usage.**
- **No phone-based, real-time alert system that warns drivers before danger.**

Why It Matters?

- **Drowsiness and distraction cause huge numbers of accidents every year.**
- **Just one moment of inattention can destroy lives and property.**
- **A smart, affordable safety solution is urgently needed.**

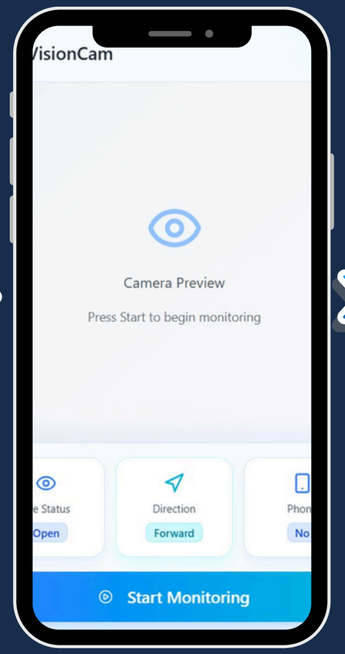
Our Solution: VisionCam

UI LAYOUT

Demo Video:https://tinyurl.com/4h6fya22

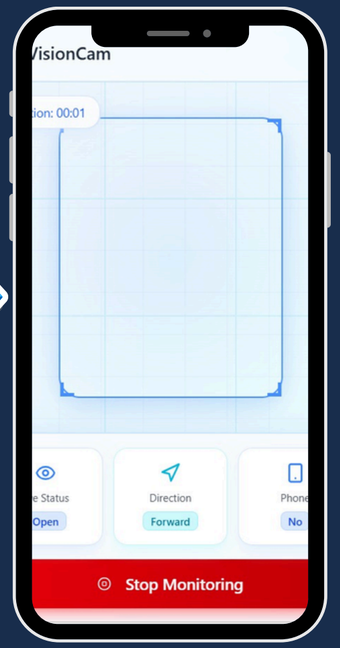
HOME PAGE

Guides the user to start monitoring with a single tap



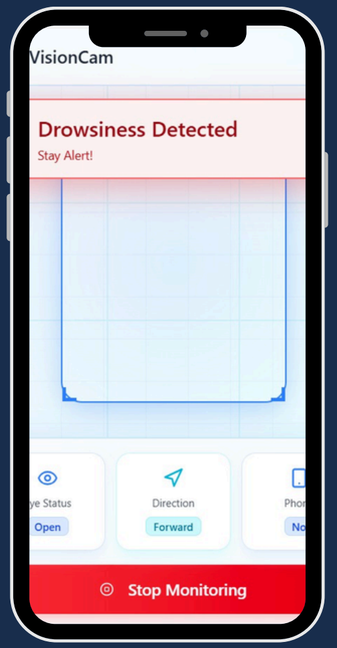
Camera Access & Setup

On enabling access, VisionCam starts capturing the driver's face in real time for analysis



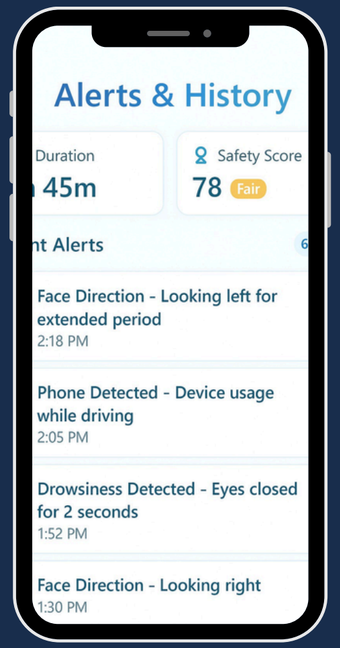
Live Monitoring Screen

AI continuously tracks facial landmarks to detect signs.



Alert Detection Screen

When unsafe behavior is detected,an instant alert is shown and alarm sounds.



Alerts & History Dashboard

This screen shows a summary of driving activity and sends alert to family or emergency contacts.

TechStack And Architecture

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FrontEnd-

- React
- Tailwind CSS
- Webcam API (live video feed)

AI Layer-

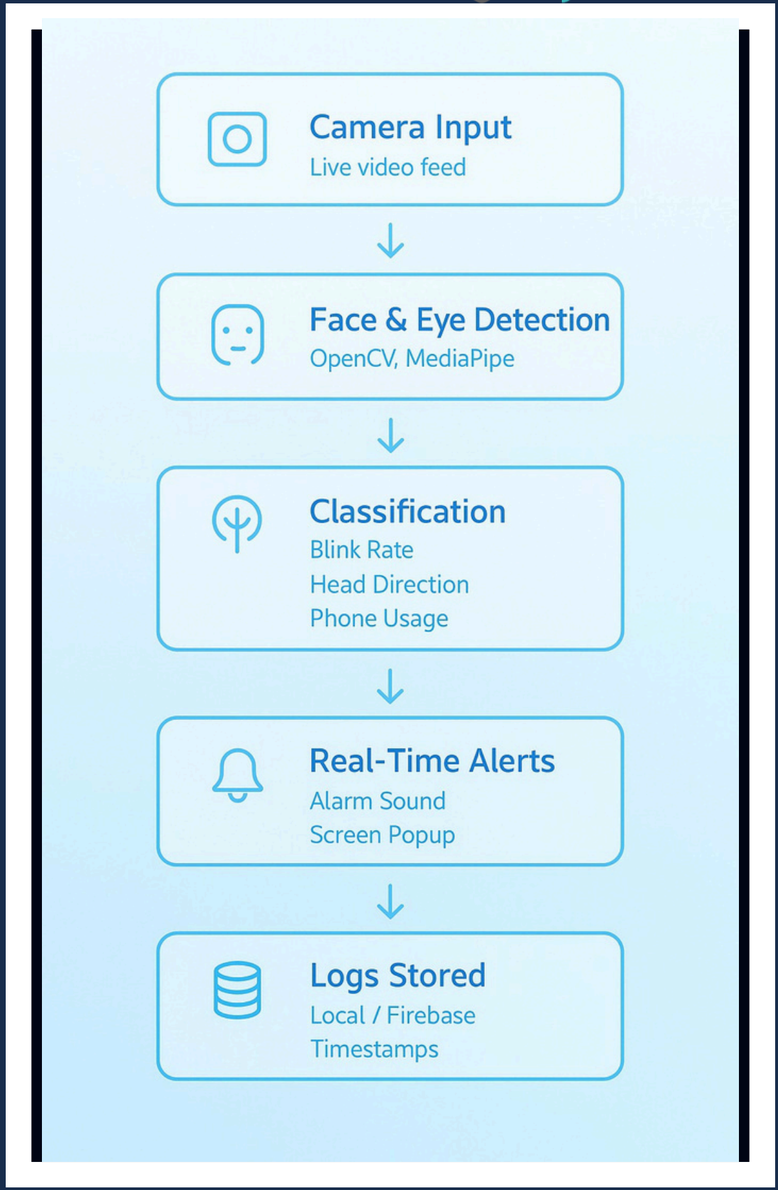
- OpenCV
- MediaPipe
- Custom alert logic

Backend-

- MongoDB
- Node.js API

Deployment-

- Netlify



USPs and Innovation

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Family Alerts

Automatically notifies family or helplines when the driver shows signs of danger or doesn't react to warnings.

Instant Alerts

Regional voice warnings immediately.

Works Anytime

Reliable even at night or in low-light conditions

No Extra Hardware

Runs on any smartphone, removing the need for expensive hardware or premium vehicles.

Offline Support

Works even on low-network highways with an offline mode for uninterrupted safety.

Behaviour Insights

Offers driver behaviour analytics, identifying risky patterns and giving personalized feedback.

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ThankYou